

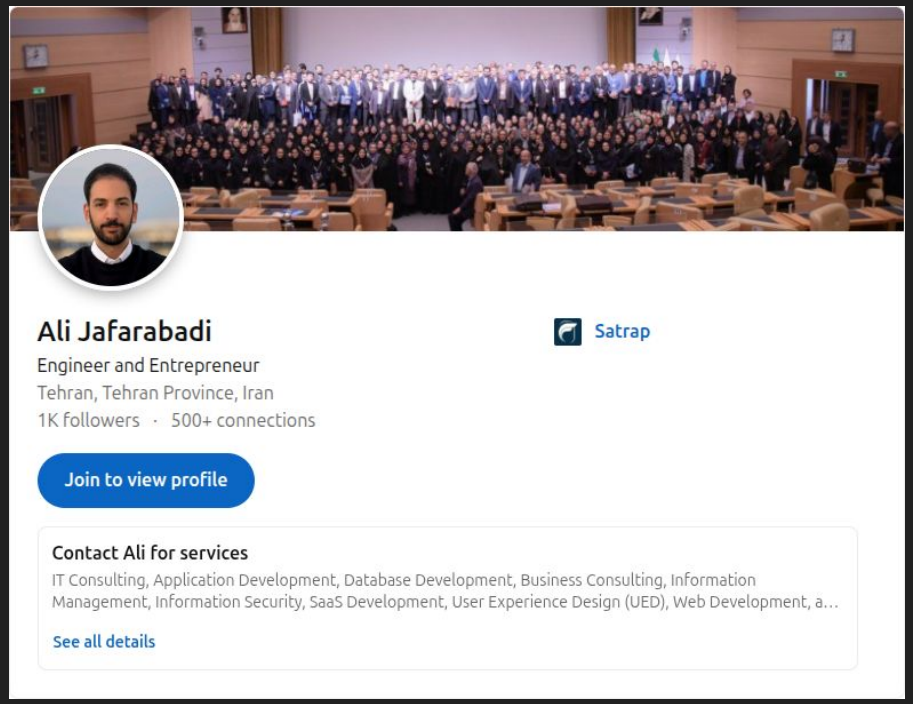


When Software Cannot Fail

“Navigating Trade-offs in Life-Critical Systems”

October 2026 - 15th Linux Fest, Amirkabir University of Technology

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  - Iran's Aviation Industry
  - Securities and Exchange Organization of Iran
  - Geographical Organization of Iran





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Safety, availability, and speed pull decisions in different directions…

A reboot is not always safe

Safety-critical

Mission-critical

Business-critical

The word “**critical**” is incomplete until we name the **consequence**.


رضا پهلوی بچه مولوی

Linux is in the room

Trade-offs

- Fail-safe ⇔ Fail-operational
- Redundancy ⇔ Complexity
- Timing ⇔ Maintainability
- Open source ⇔ Auditability
- Security update ⇔ Validated baseline
- Safety ↔ Availability
- Performance ↔ Verification
- Simplicity ↔ Features
- Automation ↔ Oversight
- …

A hazard is not a bug



A potential source of harm

People or environment are exposed to that hazard.

Injury or damage to health actually occurs.

A bug matters only if it completes this chain.

Class follows severity

IEC 62304:2006+AMD1:2015 — class is traced to the ISO 14971 risk file.

- Class A
  No injury possible
- Class B
  Non-serious injury possible
- Class C
  Death or serious injury possible

Probability does not lower the class

IEC 62304:2006+AMD1:2015

SOUP

Software of Unknown Provenance

One traceable line connects hazard to control

Hazard → software class → risk control

Thank you for taking the time;

Ali Jafarabadi (@alij8i)